

Lesson 27 — A Program That Shows Our Data

Matches LEGO Teacher Portal lesson: "Count and Crunch".

Objective (I can...): I can change a program so it collects data and shows it on screen. **CS concept:** Modifying a program that collects and visualizes data · **Builds on:** Lesson 26 (collecting and describing data).

AI Connection (2 min, read aloud): AI people make charts of their data to see it clearly — a picture of the numbers helps everyone understand. *Ask the group:* "Is it easier to understand a big list of numbers, or a picture of them?"

Materials: LEGO CS & AI kit + device per pair, the AI & Data model (from L26), color cards. **Model reuse:** AI & Data model from Lesson 26 — mostly code.

60-minute flow - Warm-up (10): "Make it a picture" — pairs turn yesterday's tally into a quick picture (stack of stickers or blocks per color). A picture of data is easier to read. - **Concept + demo (8):** Show a starter program that collects color readings and shows a simple on-screen count or bar. Change one thing (add a color, change the display) and run so kids see the chart update. (*Try "guess what it does."*) -

Build/Code (25): Pairs take the starter data program and **modify** it — add a color to collect, change what the screen shows, or add a light that flashes for the biggest group — then run it to collect and show their data. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Collect two rounds of data and compare the two charts: did the biggest group change? - **Share + tidy-up (10):** Pairs show their program collecting data and the on-screen picture of it, and read the chart aloud. Model stays built.

Pair roles: Driver modifies the program, Navigator feeds color cards and reads the chart. Swap at the wiggle break. (*Trio: third records the two rounds.*)

Differentiation: *Support* — change just one thing in the starter program. *Extension / fast-finisher* — add a third category and a label. *Reteach* — change one block, run, and see the chart move before changing the next.

Exit check: Pair modifies a program so it collects data and shows it on screen, and can read the chart.

At-home (optional): Draw a simple bar chart of something you counted (books on a shelf by color). ~5 min.